

Acoustic Emission and Passive Acoustic Regression to the ISO 281 Viscosity Ratio κ for Online, Non-Invasive Lubrication Condition Monitoring of Ball Bearings: A Controlled Dual-Channel Study

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Abstract

Inadequate lubrication is responsible for an estimated 30–80% of rolling element bearing failures, yet reliable online, non-invasive lubrication condition monitoring (LCM) remains an unsolved engineering problem. Passive sensor-based approaches are most practical for industrial retrofit because they require no bearing modification, but the signal processing challenge of separating lubrication-relevant information from confounded operating-condition effects has not been systematically characterised for simultaneous acoustic emission (AE) and passive ultrasound (US) sensing. This paper presents a controlled study in which AE and passive US signals are simultaneously acquired from a ball bearing during a continuous speed sweep (1–5985 rpm) at temperatures spanning 33–100 °C, covering the boundary, mixed, and full-film lubrication regimes across 8 783 sweep–sensor pairs per channel. The ISO 281 viscosity ratio $\kappa \in [0.003, 1.64]$ (mean 0.51) is derived analytically from bearing geometry, lubricant properties, and operating conditions as a continuous, physics-grounded regression target. For the AE channel, 26 time-domain and frequency-domain features are extracted from the broadband signal and from three frequency sub-bands (20–500 kHz, 500 kHz–1 MHz, 1–2 MHz), yielding 104 candidate features; 26 features are extracted from the pre-filtered US channel (0–20 kHz). A two-stage selection retains 32 AE and 6 US features after a joint Spearman–Pearson correlation threshold filter and a greedy variance-inflation-factor and pairwise-correlation redundancy reduction. Three regression model families are benchmarked across three sensor configurations using repeated nested five-fold cross-validation ($R = 5$ repetitions, 25 outer scores per model); LightGBM on AE features alone achieves the best hold-out performance ($R^2 = 0.951$, RMSE = 0.078 κ -units). Fusing AE and US features yields a marginal improvement that is statistically significant under repeated nested cross-validation (Nadeau–Bengio corrected t -test $p = 0.024$) but is not confirmed on the independent hold-out set (Δ RMSE = 0.0011; Wilcoxon $p = 0.88$, Diebold–Mariano $p = 0.28$, bootstrap 95% CI includes zero). Significance of pairwise model differences is assessed by Nadeau–Bengio corrected repeated k -fold t -tests with Holm–Bonferroni correction for within-sensor comparisons, and by Wilcoxon signed-rank, Diebold–Mariano, and bootstrap Δ RMSE confidence intervals for cross-sensor comparisons.

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1 Introduction

Rolling element bearings are among the most prevalent components in industrial machinery, and their reliable operation depends critically on adequate lubrication. Lubrication failure—through starvation, viscosity degradation, or contamination—increases metal-to-metal asperity contact, accelerates surface fatigue and wear, raises operating temperature, and ultimately causes premature failure. Improper lubrication is estimated to account for between 30% and 80% of all rolling element bearing failures [1–3].

Despite this, lubrication condition monitoring (LCM) has received far less attention than vibration-based fault detection and remaining-useful-life estimation, which are now mature and widely deployed [4, 5]. The lubrication signal is harder to exploit: its acoustic and vibratory signatures are weaker than the impulsive signatures of spalling or raceway cracks and are strongly entangled with speed and load, so operating-condition effects readily mask the lubrication-dependent component [5–7]. Yet accurate online LCM would enable condition-based relubrication—applying lubricant when and in the quantity actually needed, rather than on fixed schedules that systematically under- and over-lubricate [8].

The dominant bearing-monitoring approach, vibration analysis, detects macroscopic faults such as raceway spalling and rolling-element damage through characteristic-frequency analysis, but is inherently insensitive to the subtle tribological changes of film formation and breakdown [5, 9]. Temperature monitoring and oil analysis add complementary information but are indirect, subject to sampling delay, or require access to the lubricant reservoir, and none provides a direct, continuous, non-invasive indicator of the instantaneous lubrication regime under operating conditions [8].

Passive, non-invasive sensing, e.g. acoustic emission (AE) and passive ultrasound (US), is practically attractive for industrial LCM because it requires no bearing modification, no direct access to the lubricant, and can be retrofitted to bearings in service. AE captures the broadband stress waves emitted by asperity contact and microcrack events, while passive US underlies the threshold-based relubrication tools already in routine industrial use, in which the broadband ultrasonic level is compared against a baseline and relubrication is triggered once it rises by a set margin [1, 10, 11]. The physical origins of both signals are detailed in Section 2. Despite the practical maturity of threshold-based US and the demonstrated usefulness of AE, the two channels’ signal-processing behaviour has not been characterised simultaneously within a single regression framework against a physics-grounded lubrication reference.

The sensitivity of AE to lubrication regime is well established. Miettinen et al. [12] showed that AE pulse count and RMS distinguish boundary, mixed, and full-film regimes and respond to base-oil viscosity and thickener content in grease-lubricated bearings. Yoshioka and Shimizu [13] combined AE and vibration monitoring on a grease-lubricated deep-groove ball bearing using envelope demodulation and statistical AE parameters.

Tandon et al. [14] found AE and shock-pulse sensors to outperform accelerometers for detecting grease contamination in a ball bearing—the scenario closest to the present study. More recently, Cornel et al. [9] showed that AE detects lubrication-sensitive surface damage such as micro-pitting and smearing up to eight hours earlier than vibration monitoring, and Hou and Zhang [15] derived amplitude- and energy-centre features that classify lubrication status across speeds and viscosity levels, though without a continuous regression target, and with no variation of temperature.

On the signal-processing side, Renhart et al. [16] applied octave filter-bank decomposition with RMS-friction features to characterise AE tribological events, and Marticorena and García Peyrano [6] showed that spectral correlation of AE resolves lubrication-regime changes under varying load, demonstrating that kinematic and acoustic features can partially decouple operating-condition confounds. The physical link between film adequacy and the AE signature has been formalised by Liu et al. [17] and Zhang et al. [7], whose coupled vibration–AE models tie emission intensity and frequency content to EHL film thickness and the rate of asperity contact. For the passive-ultrasound channel, Jakobsen et al. [1, 11] showed that broadband acoustic features carry measurable information about the ISO 281 viscosity ratio κ .

A recurring limitation of data-driven LCM is its reliance on discrete fault labels or qualitative lubrication states rather than a physically interpretable, continuous target. The ISO 281 viscosity ratio κ removes this limitation: it is the ratio of the lubricant’s actual kinematic viscosity at operating temperature to the minimum viscosity required for full EHL film formation at the current speed and bearing geometry, mapping the instantaneous operating point to a standardised lubrication-adequacy index (Section 2). A model trained to predict κ implicitly learns the joint dependence on speed, temperature, and lubricant properties, and its output is directly interpretable against established regime boundaries. Despite this, κ has seldom been used as a regression objective in acoustic condition monitoring.

The clearest precedent is the work of Jakobsen et al. [1, 11], who established κ regression for passive bearing acoustics, predicting κ from low-cost MEMS microphone and accelerometer features using LASSO and neural networks. The present study adopts their κ target and extends it in three respects: it adds a calibrated wideband AE channel acquired simultaneously with the passive US channel, extracts frequency sub-band features from both signals, and tests whether the two modalities are complementary. This last point is an open question: AE–US complementarity is argued on physical grounds in reviews [5, 18] but has not been quantified experimentally, and it carries direct practical weight, since a second sensor adds cost and complexity and is justified only if it supplies non-redundant information. No prior study has simultaneously acquired AE and passive US from a single ball bearing and quantitatively assessed their joint value for κ regression.

This study deliberately fixes the bearing, lubricant, and load, varying only rota-

tional speed and temperature. Holding the mechanical configuration constant isolates the lubrication-dependent acoustic response from the bearing-to-bearing variability and degradation processes that would otherwise confound it—a prerequisite for establishing whether AE and passive US carry κ -relevant information at all. The resulting limits on generalisation across bearing types, lubricants, and load regimes are genuine, but not addressed here.

Contributions

This paper makes the following contributions:

1. **A dual-channel LCM dataset.** The first dataset to pair calibrated AE and passive-US signals from a single ball bearing across a continuous speed (100–3000 rpm) and temperature (40–100 °C) sweep spanning the boundary, mixed, and full-film regimes, with an analytically derived ISO 281 κ ground truth.
2. **Features of sub-bands of AE signals predict κ .** LightGBM on AE features alone reaches hold-out $R^2 = 0.951$ (RMSE = 0.078 κ -units); SHAP attribution identifies the high-frequency 1–2 MHz AE sub-band as the single most informative range for κ prediction.
3. **AE substantially outperforms passive (heterodyned) US.** The AE–US gap ($\Delta R^2 = 0.35$) is large and confirmed significant across all tests, quantifying the value of wideband AE over a heterodyned passive-US channel.
4. **The AE–US fusion question is answered quantitatively.** Combining the two channels yields only a marginal gain—statistically detectable under repeated nested cross-validation ($p = 0.024$) but not confirmed on held-out data—directly addressing the open question of whether a second acoustic modality supplies non-redundant κ information.

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The remainder of the paper is organised as follows. Section 2 reviews the theory of κ and the physical origins of AE and US signals. Section 3 describes the experimental setup and data collection. Section 4 presents the signal-processing and feature-extraction pipeline. Section 5 reports the feature analysis. Section 6 presents the regression modelling and statistical evaluation. Section 7 discusses the findings, and Section 8 concludes.

2 Background

2.1 Lubrication regimes and the viscosity ratio κ

Rolling element bearing contacts operate in one of three lubrication regimes depending on whether the lubricant film is sufficient to fully separate the rolling surfaces. In the full-

film elastohydrodynamic (EHL) regime the surfaces are completely separated and asperity contact is suppressed; in the mixed regime the film is partially formed and intermittent asperity contact occurs; in the boundary regime the film is inadequate and metal-to-metal contact dominates. The ISO 281 standard characterises these regimes through the viscosity ratio κ , with boundaries at $\kappa < 0.5$ (boundary), $0.5 \leq \kappa < 1$ (mixed), and $\kappa \geq 1$ (full film).

The ISO 281 standard defines the viscosity ratio

$$\kappa = \frac{\nu}{\nu_1(n, d_{pw})}, \quad (1)$$

where ν [$\text{mm}^2 \text{s}^{-1}$] is the actual kinematic viscosity of the lubricant at operating temperature, and ν_1 is the minimum kinematic viscosity required for full EHL at the bearing's rotational speed n [rpm] and mean pitch diameter d_{pw} [mm] [19]. The reference viscosity is given by

$$\nu_1 = \begin{cases} 45\,000 \, n^{-0.83} d_{pw}^{-0.5} & n < 1000 \text{ rpm}, \\ 4\,500 \, n^{-0.5} d_{pw}^{-0.5} & n \geq 1000 \text{ rpm}. \end{cases} \quad (2)$$

The operating viscosity ν is computed from the ASTM D341 Walther viscosity–temperature equation fitted to the lubricant's 40 °C and 100 °C reference values, with a standard value of $\lambda = 0.7$. Since ν_1 decreases with increasing rotational speed—faster rotation enhances hydrodynamic film entrainment, requiring less viscosity to achieve full EHL— κ captures the combined effect of speed, temperature, and lubricant grade on lubrication adequacy at a given operating point. Importantly, κ assumes nominally intact lubricant: the Walther equation reflects the properties of fresh grease, so κ does not encode viscosity changes arising from lubricant degradation, contamination, or starvation. Extending the regression framework to track such changes would require either continuous viscosity updates or an alternative target variable.

2.2 Physical origins of AE and passive ultrasound signals

Acoustic emission (AE) refers to transient elastic stress waves generated by sudden energy releases in a material: asperity contact and micro-fracture are the dominant sources in rolling bearings [12, 16]. Under boundary and mixed lubrication, intermittent metal-to-metal asperity contacts generate broadband stress waves that propagate through the bearing structure and are detected by piezoelectric AE sensors coupled to the outer housing [9, 16]. The emission rate, amplitude, and frequency content of these events are all sensitive to the contact severity and hence to the lubrication regime [9, 12]. Full-film EHL suppresses asperity contact, reducing AE activity; the transition through the mixed regime is associated with intermittent, structurally distinct emission events. AE sensors for bearings respond from tens of kHz into the MHz range and require correspondingly

high sampling rates to capture the full frequency content.

Passive ultrasound sensing detects structure-borne acoustic energy above the audible range. As a modality it is not intrinsically narrowband—ultrasonic emissions span from roughly 20 kHz into the MHz range, overlapping much of the AE band—but the instruments most widely deployed for bearing monitoring operate in a narrow band around 38–40 kHz. In this band the sensor amplitude is modulated by the contact-stiffness and viscous-damping changes associated with lubricant film formation: as the film degrades and asperity contact increases, the ultrasonic level at the bearing housing rises [10]. Commercial instruments (UE Systems, Sonotec) exploit this by comparing the level against a reference baseline and triggering relubrication when a defined threshold is exceeded, making threshold-based passive US the most widely deployed non-invasive LCM method in routine industrial use. The passive US probe used in this study follows the same design: it senses around 38–40 kHz and heterodynes (down-mixes) that band to an audio-frequency baseband, so the recorded US signal is band-limited to 0–20 kHz (Section 3)—a property of the available sensor rather than an inherent limit of ultrasound. However, the level is also sensitive to rotational speed and load independently of lubrication condition, which limits the reliability of threshold-based approaches under varying operating conditions.

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The two modalities thus respond to the same underlying phenomenon—increasing asperity contact as the lubricant film degrades—but through different physical mechanisms and at different frequency scales: AE captures individual high-frequency transient events (≥ 50 kHz) from discrete micro-fractures and asperity collisions, while passive US, after the probe heterodynes its 38–40 kHz sensing band to a 0–20 kHz baseband, reflects the lower-frequency bulk acoustic envelope comprising structural resonances, friction noise, and the aggregate effect of contact activity. Both channels therefore carry information about κ , and the question of whether they provide complementary or redundant lubrication-relevant information is addressed experimentally in Sections 3–6.

3 Experimental Setup

3.1 Test rig and bearing

The test rig is a custom-built bearing test stand at CeramicSpeed (Holstebro, Denmark). A single deep-groove ball bearing (model SYJ25, bore 25 mm, pitch diameter $d_{pw} = 38.0$ mm, 9 chrome steel rolling elements, dynamic load rating 14000 N) is driven by a variable-frequency electric motor controlled through a variable-frequency drive (VFD). The bearing is loaded radially by a static weight and lubricated by a defined quantity of Keratech 22 oil (kinematic viscosity $\nu_{40} = 22$ cSt, $\nu_{100} = 4.1$ cSt).

Note on VFD noise: Harmonic artefacts from the VFD switching frequency are present at approximately 5 kHz and its multiples in the low-frequency range of both sensor signals.

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3.2 Operating conditions

The experiment follows a protocol which varies the operating conditions through changes in rotational speed (rpm) and housing temperature ($^{\circ}\text{C}$). This consists of a staircase sweep from 100 to 3000 RPM and back down to 100 RPM in steps of 100 RPM with a 60-second hold at each step. This, therefore, results in This is repeated for each of 13 temperature blocks (40–100 $^{\circ}\text{C}$ in 5 $^{\circ}\text{C}$ increments) where temperature is controlled with . Sweeps of rpm are separated with a 2-minute standstill ($\text{RPM} = 0$) at the start of each block for thermal stabilisation.

The lubrication conditions in the bearing are not expected to change during the experiment in other ways than what is entailed by rpm and temperature and the effect of these upon the lubricant. It is not expected or assumed that any degradation, starvation or contamination of the lubricant happens.

3.3 Sensor configuration and data acquisition

Two sensors simultaneously acquire signals from the bearing housing through a shared oscilloscope which samples at a shared sample rate of 12.5 MHz:

Acoustic emission sensor: A wideband piezoelectric AE sensor is bonded to the bearing housing.

Ultrasound sensor: A heterodyned passive US probe is bonded to the bearing housing.

Temperature sensor:

RPM sensor:

The shared sample rate ensures that AE and US signals share timestamps.

3.4 κ derivation

The κ value for each sweep is computed according to Equations (1) and (2) using the per-sweep measured RPM and temperature, with ν evaluated via the ASTM D341 Walther viscosity–temperature equation. No separate reference measurement of film thickness is available; κ is therefore a model-derived quantity subject to the assumptions of the Hamrock–Dowson framework and Walther’s equation.

3.5 Dataset availability

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4 Signal Cleaning and Feature Extraction

The pipeline is implemented in Python using NumPy [20], SciPy [21], and antropy [22] for signal processing and feature computation, and is configured via a single YAML file to allow full reproduction from the raw HDF5 files.

4.1 Signal cleaning

The raw signals are cleaned with the following procedures:

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4.2 Filter bands

The AE and US signals are filtered with multiple low-, high- and band-pass filters, since the information on lubrication conditions contained in the signals is expected to be localised in sub-bands rather than wholly contained in the broadband signals [7, 16].

The sub-bands applied prior to feature extraction are listed in Table 1. All filters are 5th-order zero-phase Butterworth filters applied with `sosfiltfilt`.

Table 1: Frequency sub-bands used for signal pre-filtering before feature extraction.

Channel	Sub-band	Filter type	Frequency range
AE	Broadband	–	Full bandwidth
AE	Sub-band 1	Band-pass	20–500 kHz
AE	Sub-band 2	Band-pass	500 kHz–1 MHz
AE	Sub-band 3	Band-pass	1–2 MHz
US	Broadband	Low-pass	0–20 kHz

4.3 Feature extraction

Features are computed from the cleaned voltage signal for each sweep-sensor pair for each sub-band. In total 26 features are computed, comprising 12 time-domain and 14 frequency-domain statistics, as listed in Tables 2 and 3. All features (without the specific sub-bands) have previously been applied to signals for bearing condition monitoring and LCM [1, 23]; the Hjorth parameters originate from EEG analysis [24].

All frequency-domain features are computed from the one-sided FFT magnitude spectrum of the cleaned signal.

AE sub-band decomposition. In addition to broadband features, three frequency sub-bands are carved from the AE channel using zero-phase Butterworth bandpass filters: (i) 20–500 kHz, capturing the transition from audible to ultrasonic contact dynamics; (ii) 500 kHz–1 MHz, the primary stress-wave propagation band; and (iii) 1–2 MHz,

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Table 2: Time-domain features. Notation: N signal length; $\bar{x}$ sample mean; σ population standard deviation.

Name	Formula	Description
Peak	$(\max x - \min x)/2$	Half the peak-to-peak range.
RMS	$\sqrt{N^{-1} \sum_n x_n^2}$	Root mean square amplitude.
Std	$\sqrt{N^{-1} \sum_n (x_n - \bar{x})^2}$	Population standard deviation.
Variance	$N^{-1} \sum_n (x_n - \bar{x})^2$	Population variance (square of Std).
Skewness	$N^{-1} \sum_n [(x_n - \bar{x})/\sigma]^3$	Third standardised central moment; amplitude asymmetry.
Kurtosis	$N^{-1} \sum_n [(x_n - \bar{x})/\sigma]^4$	Fourth standardised central moment; tail heaviness.
Crest factor	$x_{\text{peak}}/x_{\text{RMS}}$	Peak-to-RMS ratio; sensitive to impulsive events.
Shape factor	$x_{\text{RMS}} / (N^{-1} \sum_n x_n)$	Ratio of RMS to mean absolute value.
Impulse factor	$x_{\text{peak}} / (N^{-1} \sum_n x_n)$	Ratio of peak to mean absolute value.
Margin factor	$x_{\text{peak}} / (N^{-1} \sum_n \sqrt{ x_n })^2$	Peak divided by squared mean square-root amplitude.
Hjorth mobility	$[\text{Var}(x') / \text{Var}(x)]^{1/2}$	Square root of first-derivative to signal variance ratio; proportional to mean frequency.
Hjorth complexity	$M(x')/M(x)$	Ratio of first-derivative mobility M to signal mobility; bandwidth proxy.

covering high-frequency micro-fracture events. The full 26-feature set is extracted independently for each sub-band, yielding $26 \times 4 = 104$ candidate AE features (broadband plus three sub-bands).

US channel pre-filtering. The US channel is pre-filtered to 0–20 kHz by a zero-phase low-pass Butterworth filter before broadband feature extraction, consistent with the effective bandwidth of the heterodyned probe. No frequency sub-bands are computed for the US channel, giving 26 candidate US features.

5 Feature Analysis

In this section, the correlation of features with κ and inter-feature redundancy are analysed to identify the most informative features for modelling. The procedure is as follows:

- Spearman and Pearson correlation ranking,
- Variation inflation factor calculation,
- feature selection based upon correlation and VIF,
- dimensionality reduction and visualization.

Table 3: Frequency-domain features (companion to Table 2). Notation: $S_k = |X_k|$ one-sided FFT magnitude at bin k ; f_k corresponding frequency; K bin count; $\bar{S} = K^{-1} \sum_k S_k$; $\sigma_S = \text{std}(\{S_k\})$; $\Sigma S = \sum_k S_k$; $f_c = (\Sigma S)^{-1} \sum_k f_k S_k$; $\sigma_f = (K^{-1} \sum_k (f_k - f_c)^2)^{1/2}$.

Name	Formula	Description
Dominant frequency	$f_{\arg \max_k S_k}$	Frequency at which spectral power is maximum.
Spectral mean	$\bar{S} = K^{-1} \sum_k S_k$	Arithmetic mean of the FFT magnitude spectrum.
Spectral std	$\sigma_S = \text{std}(\{S_k\})$	Population standard deviation of the FFT magnitude spectrum.
Spectral skewness	$\sum_k (f_k - f_c)^3 S_k / (\sigma_S^3 \Sigma S)$	Frequency asymmetry of the magnitude-weighted spectrum, normalised by magnitude std.
Spectral kurtosis	$\sum_k (f_k - f_c)^4 S_k / (\sigma_S^4 \Sigma S)$	Frequency tail heaviness of the magnitude-weighted spectrum, normalised by magnitude std.
Spectral flatness	$\exp(K'^{-1} \sum_{k:S_k > 0} \ln S_k) / \bar{S}$	Geometric-to-arithmetic mean ratio; 1 for white noise, 0 for a pure tone.
Centre frequency	$f_c = \sum_k f_k S_k / \Sigma S$	Magnitude-weighted mean frequency (spectral centroid).
RMS frequency	$(\sum_k f_k^2 S_k / \Sigma S)^{1/2}$	Magnitude-weighted root mean square frequency.
Freq.-weighted std	$\sigma_f = (K^{-1} \sum_k (f_k - f_c)^2)^{1/2}$	RMS deviation of bin frequencies from centre frequency (unweighted by magnitude).
f^2 -wt. freq.	$(\sum_k f_k^4 S_k / \sum_k f_k^2 S_k)^{1/2}$	RMS frequency under $f_k^2 S_k$ weighting; emphasises higher frequencies relative to RMS frequency.
Norm. freq. std	σ_f / f_c	Frequency standard deviation normalised by centre frequency.
Frequency skewness	$\sum_k (f_k - f_c)^3 S_k / (K \sigma_f^3)$	Third spectral moment of frequency deviation from centroid, weighted by magnitude.
Frequency kurtosis	$\sum_k (f_k - f_c)^4 S_k / (K \sigma_f^4)$	Fourth spectral moment of frequency deviation from centroid, weighted by magnitude.
Norm. band-width	$\sum_k f_k - f_c ^{1/2} S_k / (K \sigma_f^{1/2})$	Half-order moment of spectral spread, normalised by $\sigma_f^{1/2}$.

5.1 Correlation with κ

Table 4 lists the top ten features for each sensor ranked by mean Spearman and Pearson correlation with κ across all sweeps, and Figure 1 and Figure 2 show the full ranking of all features by absolute Spearman correlation.

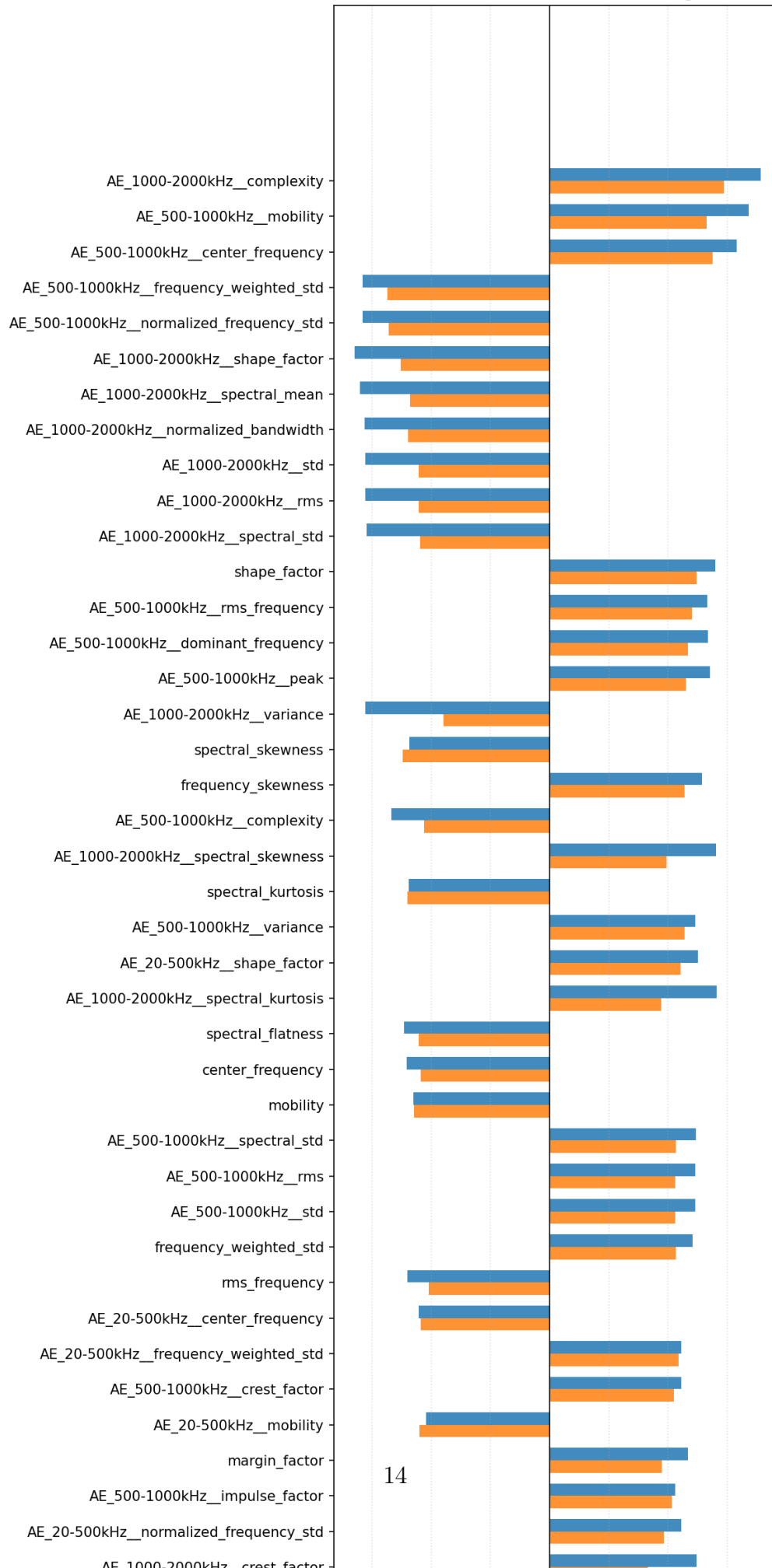
Table 4: Top 10 features by combined Spearman–Pearson rank score with κ , for the AE and the US signals.

AE			US		
Feature	$ \rho $	$ r $	Feature	$ \rho $	$ r $
1–2 MHz complexity	0.891	0.736	Spectral mean	0.663	—
500k–1 MHz mobility	0.841	0.663	Peak	0.629	—
500k–1 MHz centre freq.	0.789	0.687	RMS	0.607	—
500k–1 MHz freq.-wt. std	0.789	0.686	Std	0.606	—
500k–1 MHz norm. freq. std	0.789	0.679	Spectral std	0.607	—
1–2 MHz shape factor	0.825	0.628	Variance	0.606	—
1–2 MHz spectral mean	0.801	0.590	Norm. bandwidth	0.607	—
1–2 MHz norm. bandwidth	0.782	0.599	Freq. skewness	0.539	—
1–2 MHz std	0.777	0.552	Freq. kurtosis	0.537	—
1–2 MHz RMS	0.777	0.552	Kurtosis	0.469	—

The top AE features are drawn overwhelmingly from the 1–2 MHz and 500 kHz–1 MHz sub-bands, with sub-band complexity ($|\rho| = 0.891$) and mobility ($|\rho| = 0.841$) heading the ranking. Broadband AE features rank lower than their sub-band counterparts, indicating that the frequency-specific content of the high-frequency AE signal carries more monotonic information about κ than the full-band aggregate [7, 16]. For the US channel, amplitude features (spectral mean, peak, RMS, std) dominate at $|\rho| \approx 0.60$ – 0.66 ; the maximum US correlation ($|\rho| = 0.663$) is substantially lower than the top AE correlation ($|\rho| = 0.891$). Figure 1 and Figure 2 show the full feature correlation rankings for each sensor.

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AE — feature correlation ranking



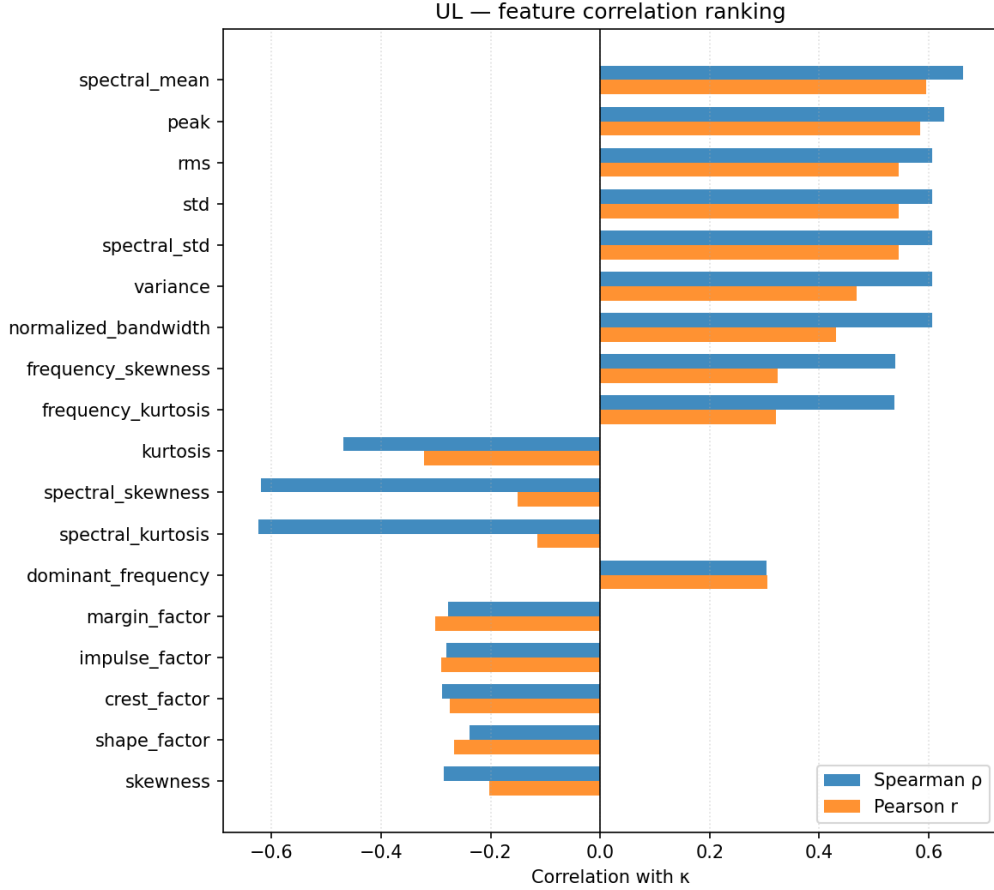


Figure 2: Spearman $|\rho|$ and Pearson $|r|$ between each US-channel feature and κ , sorted by combined rank. Retained features highlighted.

5.2 Operating-condition confounding

The raw Spearman correlation between broadband signal energy (as a proxy for all amplitude features) and κ is substantially driven by the shared dependence on rotational speed and temperature. Quantitatively, RPM has a Spearman correlation of $\rho = +0.62$ with κ and temperature has $\rho = -0.68$. Higher RPM reduces the required viscosity ν_1 (Eq. (2)) and therefore increases κ , while simultaneously increasing signal energy through more frequent contact events per second. Higher temperature reduces the actual viscosity ν and therefore reduces κ , while also affecting material damping and structural dynamics.

As a consequence, amplitude-dominated features such as RMS, variance, and peak — which rank among the top features by raw Spearman correlation — are partially acting as indirect proxies for rotational speed rather than as direct indicators of lubrication film adequacy [7, 12, 15]. This confounding is an inherent property of any single-operating-point sensor system without an explicit operating condition normalisation step.

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5.3 PCA and regime separability

Figure 3 shows the first two principal components of the retained AE and US features sets coloured by κ value. Visual inspection shows that some separability in this subspace is possible. This confirms that the retained feature sets carry regime-relevant discriminative information.

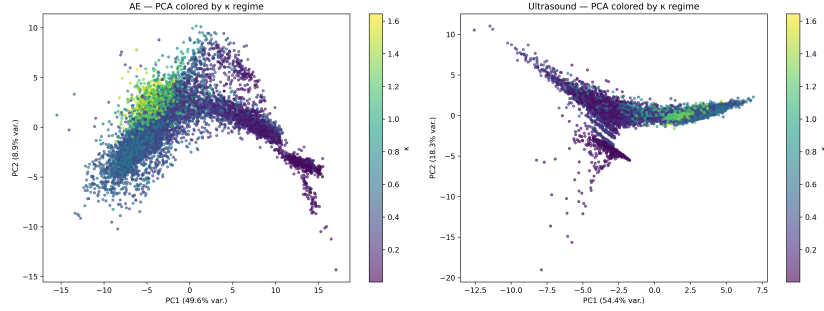


Figure 3: PCA projection of the retained AE and US feature sets, coloured by lubrication κ value.

6 Regression Modelling and Statistical Evaluation

6.1 Experimental design

6.1.1 Data sets and splits

The data experiments utilise three different features sets: the AE, the US, and the combined AE and US feature sets. Each of the later described models are fitted to each of the three datasets. Each dataset is divided into a training set (80%) and a hold-out test set (20%) by sweep-level random sampling, stratified to preserve the κ regime distribution. The hold-out set is withheld entirely from all model development steps and used only for final evaluation.

6.1.2 Model families

Three regression model families are evaluated for three sensor configurations, as summarised in Table 5.

6.2 Repeated nested cross-validation

To obtain stable performance estimates that properly account for the variance introduced by the train/validation split, the model evaluation uses a repeated nested k -fold cross-validation scheme. The outer loop is repeated $R = 5$ times with independently re-seeded five-fold splits, yielding $R \times k = 5 \times 5 = 25$ outer RMSE scores per model–sensor combination. For LightGBM, each outer fold contains its own inner Optuna optimisation loop,

Add Elastic-NetCV search grid values (number of α candidates

Model	Rationale	Hyperparameters & tuning
Elastic Net	$\ell_1 + \ell_2$ regularised linear model; ℓ_1 promotes sparsity, ℓ_2 stabilises correlated features.	α (strength) and ℓ_1 ratio; ElasticNetCV inner CV loop.
Polynomial (deg. 2)	Augments linear model with all pairwise interaction terms; remains linear in parameters.	Ridge penalty λ ; RidgeCV within each outer fold.
LightGBM	Gradient-boosted trees; captures non-linear feature- κ mappings and high-order interactions.	Learning rate, num. leaves, max. depth, min. child samples, ℓ_2 reg.; Bayesian optimisation (Optuna TPE, 20 trials per outer fold).

Table 5: Regression model families: rationale and hyperparameter tuning strategy.

ensuring that the reported outer scores are not contaminated by hyperparameter selection on the outer fold data. For the linear models, hyperparameter selection is handled by **ElasticNetCV** and **RidgeCV** within each outer training partition.

The 25 outer scores form the empirical sampling distribution of RMSE for each model and are used directly in the significance tests described below.

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6.3 Statistical significance testing

Pairwise significance tests are conducted at two levels.

Level 1 — Within-sensor comparisons (architecture differences). For pairs of models trained on the same sensor configuration (e.g., Elastic Net vs. Polynomial on AE), the corrected repeated k -fold t -test of Nadeau and Bengio [25] is applied to the 25 paired RMSE scores. This test corrects the standard paired t -test for the positive correlation induced by shared training data across repeated folds. Multiple within-sensor comparisons are adjusted by the Holm–Bonferroni step-down procedure at $\alpha = 0.05$.

Level 2 — Cross-sensor comparisons (best per sensor configuration). For the best model from each sensor configuration (selected by lowest mean CV RMSE), pairwise differences on the common hold-out set are assessed by:

- The Wilcoxon signed-rank test on paired residuals (non-parametric);
- The modified Diebold–Mariano test for equal predictive accuracy under squared-error loss;
- A bootstrap Δ RMSE 95% confidence interval (10 000 resamples with replacement).

Cross-sensor p -values are also corrected by Holm–Bonferroni.

6.4 Results

Model	Sensors	CV RMSE	HO R^2	HO MAE	HO RMSE
LightGBM	Combined	0.0767 ± 0.0040	0.953	0.050	0.076
LightGBM	AE	0.0780 ± 0.0037	0.951	0.051	0.078
LightGBM	US	0.2233 ± 0.0060	0.600	0.155	0.222
Elastic Net	Combined	0.1556 ± 0.0051	0.830	0.111	0.145
Elastic Net	AE	0.1568 ± 0.0052	0.826	0.113	0.147
Polynomial	Combined	0.1568 ± 0.0048	0.803	0.113	0.156
Polynomial	AE	0.1646 ± 0.0044	0.775	0.123	0.167
Polynomial	US	0.2577 ± 0.0056	0.477	0.193	0.254
Elastic Net	US	0.2783 ± 0.0068	0.396	0.210	0.273

Table 6: Regression model performance. CV RMSE: mean $\pm$ std of outer RMSE across 25 repeated nested CV folds ($n_{\text{train}} = 7\,026$ per fold). HO: hold-out test set ($n = 1\,757$). Best hold-out R^2 in bold.

Overall performance. LightGBM achieves the best hold-out performance across all sensor configurations, with AE-only reaching $R^2 = 0.951$, RMSE = 0.078 κ -units. On a target range of [0.003, 1.64] with mean 0.51, this corresponds to a relative RMSE of approximately 15% of the target range. Linear and polynomial models on AE features achieve $R^2 \approx 0.77$ –0.83, confirming a non-linear feature- κ mapping that gradient boosting exploits more effectively.

Sensor comparison. AE features substantially outperform US features across all model families. The LightGBM AE-US gap ($\Delta R^2 = 0.35$) is confirmed as highly significant (Nadeau-Bengio $p < 10^{-22}$; Wilcoxon $p < 10^{-166}$; $\Delta\text{RMSE} = 0.144$, 95% CI [0.133, 0.157]). This reflects the richer frequency content and more direct physical sensitivity of the AE sensor to asperity contact events.

Sensor fusion yields a marginal, statistically significant. LightGBM on the combined AE+US feature set achieves $R^2 = 0.953$, marginally above the AE-only result (0.951). The within-CV corrected t -test rejects equality ($p = 0.024$), but hold-out tests do not confirm the improvement: Wilcoxon $p = 0.88$, Diebold-Mariano $p = 0.28$, bootstrap $\Delta\text{RMSE} = 0.0011$ with 95% CI [−0.0009, 0.0033] spanning zero. For the linear models, the combined feature set is similarly on par with AE alone (Elastic Net: R^2 0.830 vs. 0.826; Polynomial: 0.803 vs. 0.775). The marginal fusion gain is consistent with the US channel contributing a small amount of non-redundant information at low-frequency.

Model consistency. The close agreement between nested CV RMSE and hold-out RMSE across all models indicates stable generalisation with no evidence of significant overfitting.

Figure 4 shows predicted vs. actual κ for the hold-out set for all model families under the AE-only configuration.

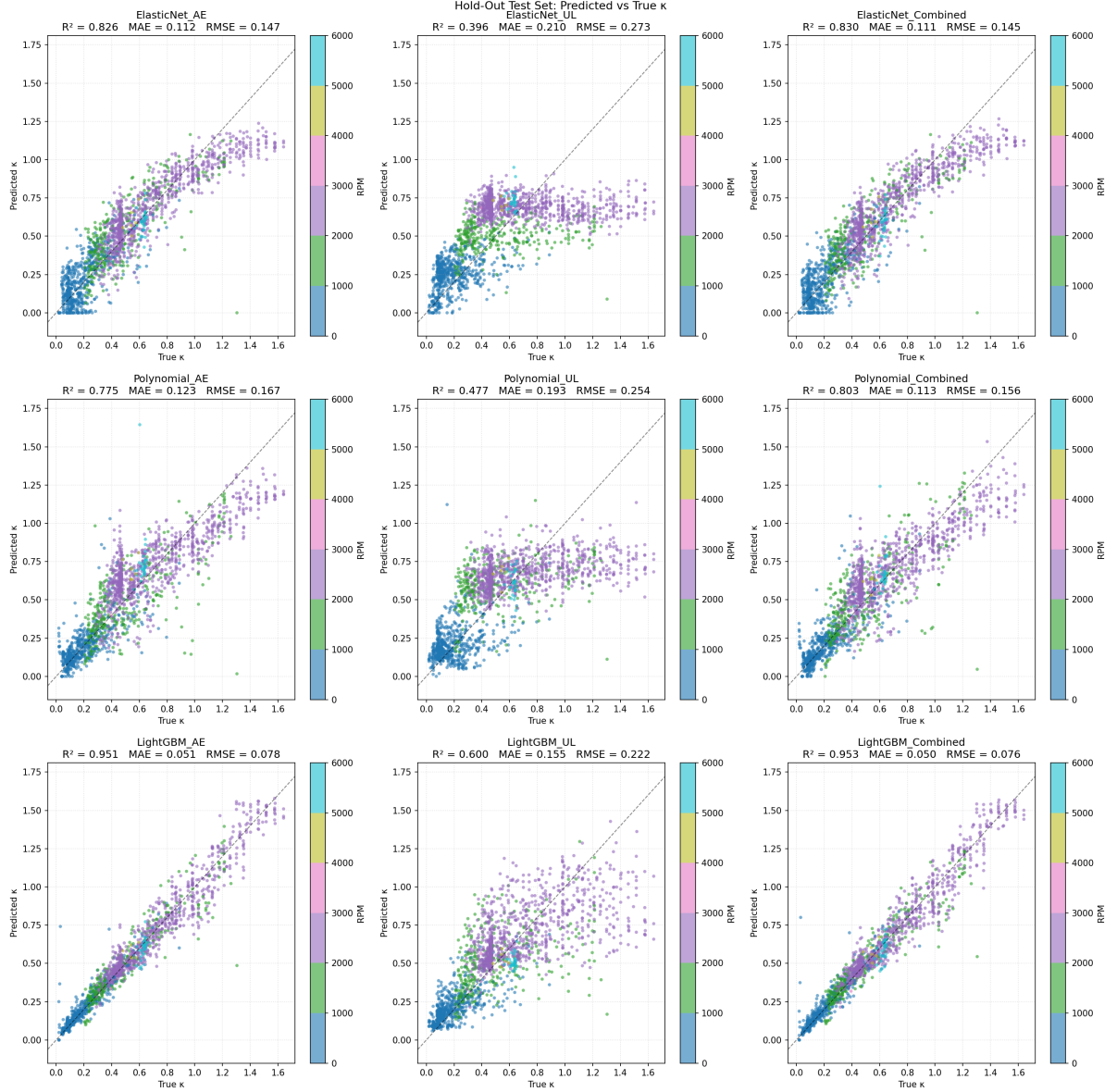


Figure 4: Predicted vs. actual κ on the hold-out test set for all model families and sensor configurations, colored according to RPM. The dashed diagonal indicates perfect prediction. Residual spread increases in the mixed lubrication regime ($0.5 \leq \kappa < 1.0$).

SHAP-based feature importance. SHAP (SHapley Additive exPlanations) values computed via TreeExplainer identify which AE sub-band features drive the κ predictions. The dominant contributor by a large margin is *1–2 MHz complexity* (mean $|\text{SHAP}| = 0.162$), followed by *500 kHz–1 MHz centre frequency* (0.051) and *1–2 MHz mobility* (0.038). The 1–2 MHz band complexity feature is also the only feature ranked in the top 10 across all three model types (cross-model agreement, Table 7), making it the most robust single predictor of κ in the dataset. Figure 5 shows the full importance ranking.

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Feature	ElasticNet	Polynomial	LightGBM	Models in top 10
1–2 MHz complexity	1	1	1	3
1–2 MHz mobility	2	—	3	2
500k–1 MHz spectral std	3	—	10	2
Broadband freq. skewness	10	—	4	2
500k–1 MHz centre freq.	—	10	2	2
500k–1 MHz dominant freq.	—	4	7	2

Table 7: Cross-model SHAP agreement for AE features: features appearing in the top 10 by mean $|\text{SHAP}|$ for more than one model type.

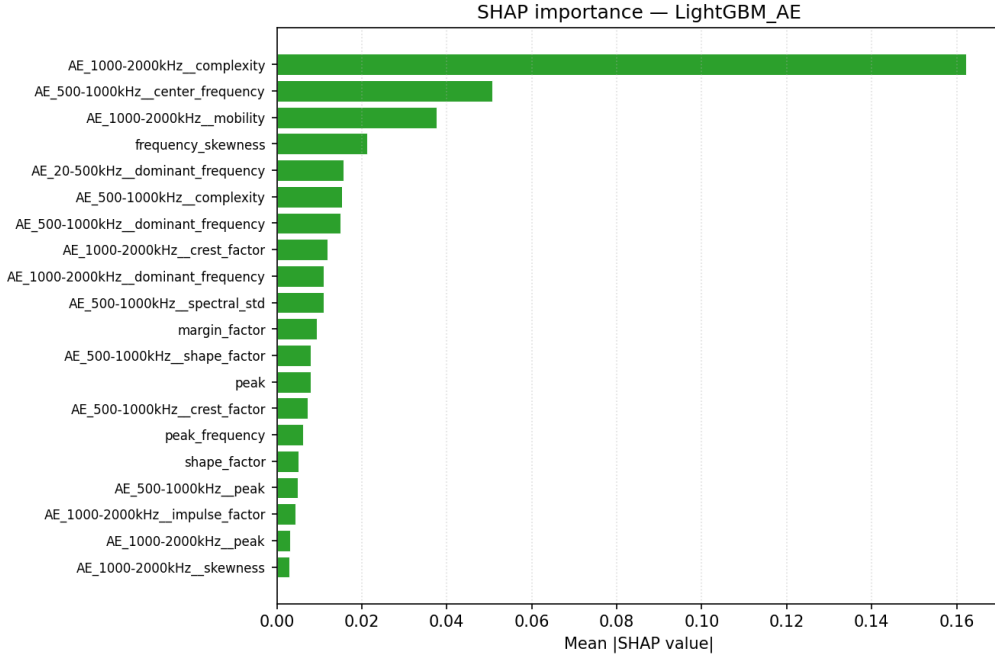


Figure 5: SHAP feature importance (mean $|\text{SHAP value}|$) for the LightGBM model trained on AE features. Features from frequency sub-bands are prefixed with the band label (e.g., AE_20–500kHz_).

7 Discussion

7.1 AE vs. US

The consistently superior performance of AE over the US channel across all model families reflects a fundamental hardware difference. The AE sensor captures genuine wideband stress-wave emissions extending into the MHz range (analysed here up to 2 MHz), directly sensitive to individual asperity contact events. The US channel is heterodyned, limiting its useful content to below approximately 10 kHz, where the signal is a mixture of low-frequency machinery noise, VFD harmonics, and structural vibration. The physical information about microscale contact events is largely absent in this frequency range.

Replacing the heterodyned probe with a wideband passive acoustic or ultrasound sensor would be expected to substantially improve the US-channel models.

7.2 Fusion gain

The marginal fusion result ($\Delta R^2 = 0.002$ on the hold-out, not confirmed by Wilcoxon or Diebold–Mariano tests) can be understood from two complementary angles. First, the US channel retains only 6 features after selection (vs. 32 for AE), and the top US correlations ($|\rho| \leq 0.66$) are far weaker than the top AE correlations ($|\rho| = 0.89$). The US channel therefore contributes a small amount of additional information that provides a marginal benefit to the combined model under CV, but the effect is too small to be detectable on the hold-out set (1757 samples). Second, the US channel’s effective bandwidth (< 10 kHz after heterodyning) means its information content partially overlaps with the low-frequency tail of the AE signal, limiting the independent contribution it can make.

This result is an informative contribution to the LCM literature, where AE–ultrasound complementarity is widely *argued* but rarely *tested*. Whether a wideband passive ultrasound sensor (rather than a heterodyned probe) would provide more independent information – and thus a more substantial fusion gain – remains an open question for future work.

7.3 Interpreting the amplitude-confounding result

The strong raw Spearman correlations of amplitude features (RMS, variance, Hjorth mobility) with κ should be interpreted with caution. As discussed in Section 5.2, rotational speed is a driver of both κ (through ν_1 in Eq. (2)) and broadband signal energy (through the rate of and energy in contact events per revolution as described in [7]). A model trained on amplitude features therefore exploits speed as an indirect proxy for κ rather than sensing purely lubrication-specific signal changes directly.

This confounding does not invalidate the regression results: a model that uses speed-mediated amplitude changes to predict κ is still predicting κ , and its predictions may be operationally useful. However, it does indicate that the model’s generalisation to operating conditions outside its training distribution (different speed ranges, different bearing loads) may be limited. Future feature engineering should prioritise spectral shape and ratio features (frequency kurtosis, normalised bandwidth, spectral flatness) that are expected to carry a higher proportion of genuine lubrication-state information relative to operating condition proxies.

7.4 Implications for the κ regression framework

The κ regression framework is validated as a methodologically sound approach for passive non-invasive LCM, when κ is assumed to be representable of the lubrication conditions in the bearing. A hold-out $R^2 = 0.951$ and $\text{RMSE} = 0.078$ κ -units for a single AE sensor represents strong performance across a demanding continuous speed-and-temperature ramp spanning boundary through full-film lubrication conditions. This exceeds the performance reported by Jakobsen et al. [1] in a comparable passive sensing study with US.

The repeated nested cross-validation design provides appropriately conservative performance estimates: the 25 outer RMSE scores capture fold-to-fold variance without the optimistic bias introduced by simple k -fold evaluation with shared hyperparameter search. The corrected significance tests confirm that performance differences between model families are assessed under assumptions that properly account for data re-use.

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7.5 Limitations

1. **Single bearing and lubricant.** All results are specific to the SYJ25 bearing and Keratech 22 oil. Generalisability to other bearing types, sizes, or lubricants is unknown.
2. **Heterodyned US channel.** The passive US channel is frequency-limited to ≈ 20 kHz. It should not be equated with wideband ultrasound and results should be interpreted in light of this hardware constraint.
3. **Data gap.**
4. **Model-derived κ reference.**

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8 Conclusion

This paper has presented a controlled study of simultaneous acoustic emission and passive ultrasound sensing regressed to the ISO 281 viscosity ratio κ for non-invasive lubrication condition monitoring of a ball bearing. The main findings are as follows:

1. **AE sub-band features are strong predictors of κ .** LightGBM on 32 retained AE features achieves hold-out $R^2 = 0.951$, $\text{RMSE} = 0.078$ κ -units. The 1–2 MHz complexity feature is the dominant SHAP contributor and the only feature consistent across all three model types, identifying the high-frequency AE sub-band as the most informative frequency range for lubrication regime prediction.

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2. **Sensor fusion yields a marginal gain—significant under cross-validation, unconfirmed on hold-out.** The LightGBM combined model achieves $R^2 = 0.953$ ($\Delta R^2 = +0.002$ over AE alone); the within-CV corrected t -test rejects equality ($p = 0.024$), but hold-out significance tests (Wilcoxon $p = 0.88$, Diebold–Mariano $p = 0.28$, bootstrap 95% CI including zero) cannot confirm the improvement. The heterodyned US channel contributes only 6 retained features with substantially weaker correlations than AE, limiting its independent contribution.
3. **AE strongly and significantly outperforms the US channel.** The AE–US gap ($\Delta R^2 = 0.35$ for LightGBM) is confirmed highly significant by all test statistics, reflecting the fundamental bandwidth limitation of the heterodyned US probe.
4. **Rigorous evaluation via repeated nested cross-validation.** Twenty-five outer RMSE scores per model provide stable, conservative performance estimates; corrected significance tests confirm that LightGBM significantly outperforms linear models within each sensor configuration, while the Elastic Net–Polynomial difference is not significant for the combined feature set.
5. **The κ regression framework is validated** as a highly effective approach for passive non-invasive LCM, achieving substantially higher performance than prior comparable studies, attributed to the sub-band feature set and the large continuous-sweep dataset.

Future work should address the identified limitations by: acquiring wideband passive ultrasound to replace the heterodyned probe; validating across multiple independent sessions, bearing types, and lubricants; investigating temporal autocorrelation and its effect on hold-out validity; and developing physically motivated sub-band ratio features that are explicitly designed to be robust to operating-condition variation.

A Full Feature Correlation Rankings

Tables ?? and ?? list the complete Spearman and Pearson correlations with κ for all AE and US features that passed the initial data quality filter, sorted by combined rank. Features retained after the Stage 2 VIF and pairwise-correlation redundancy reduction are indicated with a checkmark.

Run scripts/02_feature_analysis.py to generate
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Run scripts/02_feature_analysis.py to generate
feature_ranking_ul_appendix.tex.

CRediT authorship contribution statement

Frederik Appel Vardinghus-Nielsen: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interests

The author declares that there are no competing interests.

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IFD?

Data availability

The complete dataset (HDF5 signal files, extracted features in Parquet format, and the reproducible Python analysis pipeline) will be deposited on Zenodo upon acceptance [DOI to be assigned].

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used AI-assisted writing tools for language editing and structural drafting. After using these tools, the author reviewed and edited the content as necessary and takes full responsibility for the content of the publication.

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